



Machine Learning and Data Science for Policy Analysis

Fall Quarter 2026

Mondays and Wednesdays 11:30am-1pm ET, 8:30am-10am PT

Professors Carter Price and Gabe Hassler

Professors' Office Hours: Fridays 12-1pm ET, 9-10am PT

Course Description

This 10-week course will introduce students to foundational concepts in Machine Learning (ML) practice, building on their prior exposure to probability and statistical concepts. The class will cover the theoretical underpinnings of machine learning practices but focus on practical uses. Students with less quantitative experience will gain exposure to the mathematics behind the basic methods. Students with a significant quantitative background will explore the full life cycle of a data science project and best practices.

Course Learning Outcomes

Upon completion of this course, a student should understand:

- The workflow for an ML project,
- The basic mechanics of key ML techniques,
- How these methods fail, and
- How to be a critical consumer of ML analysis.

Additionally, through the homework assignments, the student should know how to use AI-assisted programming tools responsibly by specifying tasks clearly, inspecting generated code, testing outputs, documenting use, and retaining responsibility for all submitted work.

Finally, the student should be able to assess whether an AI system is (in)appropriate for a particular policy decision.

Prerequisites

- Math skills: Basic to Intermediate Calculus, Basic optimization concepts: Linear algebra, derivatives, integrals, basic multivariable calculus.
- Introductory Probability and Statistics: Probability distributions, statistical concepts (e.g., Law of Large Numbers /Central Limit Theorem/Confidence Intervals)
- Basic Statistical Modelling: Basic regressions (linear and logit)
- Programming skills: Basic ability to manipulate data in Python, R, Julia, or a similar programming language

Auditors

Auditors are welcome, up to a limit set by the School.

Class Modality

The modality for this course is Hybrid. There will be two class sessions per week. Mondays will be in-person at the SM campus, and Wednesdays will be in-person at the DC campus. DC-based students will attend the SM-based class via Teams, and vice versa.

Note: all students are expected to complete the midterm and final in person. If the midterm is on a Monday, for example, both DC and SM students are expected to be in person on that day at their respective campus.

- Students at the Campus Location: The student is required to attend all class sessions in person at the assigned campus.
- Non-Campus Students: If the student is not assigned to the campus location, the student is expected to attend all class sessions virtually via Teams.
- If a student is unable to attend class for personal reasons, they must notify the professor at least **two business days prior to the class session**. The student is responsible for contacting the professor to discuss how to obtain class materials and make up any missed work.

Attendance

Students are expected to attend all class sessions in person or via Teams. All students who are unable to attend a class session must notify the instructor as soon as practicable after the inability arises.

Evaluation Criteria

- 10% Class Participation
- 30% Homework
- 10% Quizzes
- 25% Midterm
- 25% Final

Submitting Work Late

Late submission of work is not permitted, and will result in a half-letter grade reduction for each day late. In extraordinary and unforeseeable circumstances (extreme family emergency, severe health crisis), and **with prior permission** from the instructor, late submission may be authorized.

Student Workload

As a rule of thumb, a student unfamiliar with course content will need to spend at least two hours of out-of-class work for each hour of coursework, i.e., 6 hours of out-of-class work each week. A student's actual time requirement will naturally depend on many factors, and rigor is measured primarily by a course's learning objects along with evidence of student achievement.

Use of AI Tools

Generative-AI tools can be useful for programming, debugging, drafting, brainstorming, and exploring technical concepts. They can also produce incorrect code, fabricated citations, misleading explanations, insecure practices, and analyses that appear plausible but are invalid. Students may use approved AI tools as documented in course assignments, but they remain fully responsible for the accuracy, originality, reproducibility, and ethical implications of all submitted work.

For this course,

- Generally permitted with disclosure:
 - Explaining concepts or error messages.
 - Generating initial code, pseudocode, tests, documentation, comments, or visualizations.
 - Debugging code.
 - Questioning data dictionaries.
 - Editing prose for clarity, grammar, or structure.
 - Summarizing course readings when the student verifies against the original source.
- Permitted only with explicit disclosure and verification:
 - Generating interpretation of model results.
 - Identifying sources, citations, or factual claims.
- Prohibited
 - Completing the AI use disclosure form.
 - Submitting AI-generated work that the student cannot explain, reproduce, or defend.
 - Using AI tools during quizzes or examinations unless explicitly authorized.
 - Fabricating sources, data, code outputs, model results, or citations.
 - Delegating the full intellectual work of an assignment to an AI system.

As part of each assignment, there will be an AI disclosure form where the student describes how AI was used, which tools were used, and, in some cases, the specific prompts or responses.

Communicating with the Instructor

Students can communicate with professors during office hours or via email or Teams. Questions about class content or assignments should be reserved for office hours. Office hours will be Fridays 12pm-1pm ET, 9am-10am PT via Teams.

Electronic Devices in Class

Classes will be hosted on Teams. Discussions are under Chatham House rules. Please do not record the class without permission. You are permitted to use any nonintrusive electronic device that supports your learning. **Electronic devices will *not* be used for nonacademic purposes during lecture/discussion in class.**

Class Meeting Schedule and Topics

Each class will have the following structure:

- Lectures of 50 minutes each
 - Using R or Jupyter notebooks for examples and illustration
 - PPT Slides for delivery
- Last 25 minutes:
 - Free-form QA
 - Discussion of Readings
 - Short weekly quizzes meant to reinforce key concepts

Homework with quantitative exercises and essays: There will be weekly computational assignments as well as less frequent conceptual assignments. While computational assignments may be completed using any programming language of your choice, students are expected to be able to program in at least one relevant language (e.g., Python, R, Julia). Computational assignments will be submitted via GitHub, while conceptual assignments will be turned in via email.

COURSE SCHEDULE		
Week	Topic	Homework
Week 1	Overview and Key Concepts • Background concepts • ML workflow • Pitfalls and limitations <i>Additional Practical Topics</i> • <i>Getting started with Python</i> • <i>Integrated development environments (IDEs)</i>	• Short Essays on failure modes • Math exercises • Set up relevant accounts (e.g. GitHub)
Week 2	Data Management • Acquisition • Processing • Transformation • Standards	• Exercises and code reviewing data formatting • Debugging LLM generated code

	• Visualization <i>Additional Practical Topics</i> • <i>Git and version control</i> • <i>GitHub and code sharing</i>	
Week 3	Evaluation • Error measures • Training and testing • Cross validation Optimization for AI/ML • Analytic • Numerical <i>Additional Practical Topics</i> • <i>Introduction to AI programming assistants for ML workflows</i>	• Exercises and code reviewing error measures and cross-validation • Exercises and code on formulating optimizations
Weeks 4-5	Unsupervised Learning • Evaluation • Principal Component Analysis • K-means • Agglomerative Hierarchical Clustering • Visualizations <i>Additional Practical Topics</i> • <i>How to get the most out of AI programming assistants for ML workflows</i>	• Code for various methods and the goals **Midterm in-class**
Weeks 6-7	Supervised Learning • Evaluation approaches • Classification and Regression Trees • Random Forests (time permitting) • Neural Networks • Visualizations <i>Additional Practical Topics</i> • <i>Data security</i>	• Code for solving a data analysis task (with clear evaluation workflows identified)
Week 8	Reinforcement Learning • Reward functions • Design • Failure modes <i>Additional Practical Topics</i> • <i>Selected topics in ML computing</i>	• Code for solving a data analysis task (with clear evaluation workflows identified)
Week 9	End-to-end evaluation • Troubleshooting • Evaluation approaches <i>Additional Practical Topics</i> • <i>Selected topics in ML computing</i>	• Essay on evaluation

Week 10	Bonus Topics • Computer vision • Machine translation <i>Additional Practical Topics</i> • <i>Reproducible research</i>	**Final In-Class**
---------	---	---------------------------

Course Readings

The primary textbook for the course is:

Géron, Aurélien. *Hands-On Machine Learning with Scikit-Learn and PyTorch*. O'Reilly. 2025.

<https://www.oreilly.com/library/view/hands-on-machine-learning/9798341607972/>

Note that this course will cover the first half of the book. This is also the text book for the Neural Networks course, which covers the second half of the book.

Overview and Key Concepts

Price, Carter C., Sandler, Morgan. Automating Knowledge Work: A Framework for Identifying Automation Strategies for Tasks. In: Rangeley, M., Fairfax, N. (eds) *The Artificial Intelligence Revolution*. Springer, Cham. 2026. (on course website)

Geron, Chapter 1

Nisbet, Elder, and Miner. “Top 10 Data Mining Mistakes” [on course website]

Data Management

Turner Lee, Resnick, and Barton “Algorithmic bias detection and mitigation: Best practices and policies to reduce consumer harms” [available [here](#)]

Geron, Chapter 2

Optimization

No reading

Unsupervised Learning

Geron, Chapter 3, 7, and 8

Anderson, Tessa K. “Kernel Density Estimation and K-Means Clustering to Profile Road Accident Hotspots.” *Accident Analysis & Prevention*, vol. 41, no. 3, May 2009, pp. 359–64, <https://doi.org/10.1016/j.aap.2008.12.014>.

Richmond, Craig S., et al. “SARS-CoV-2 Sequencing Reveals Rapid Transmission from College Student Clusters Resulting in Morbidity and Deaths in Vulnerable Populations.”

MedRxiv, Oct. 2020, p. 2020.10.12.20210294,
<https://doi.org/10.1101/2020.10.12.20210294>.

Supervised Learning

Geron, Chapters 4 (skim), 5, 6, and 9

Gadaleta, Matteo, et al. "Passive Detection of COVID-19 with Wearable Sensors and Explainable Machine Learning Algorithms." MedRxiv, Aug. 2021, p. 2021.08.05.21261626,
<https://doi.org/10.1101/2021.08.05.21261626>.

Reinforcement Learning

Geron, Chapter 19 (skim)

End-to-End Evaluation

Wells Fargo Mortgage Bias:

Shawn Donnan, Ann Choi, Hannah Levitt and Christopher Cannon. "Wells Fargo Rejected Half Its Black Applicants in Mortgage Refinancing Boom," Bloomberg. March 11th, 2022.
<https://www.bloomberg.com/graphics/2022-wells-fargo-black-home-loan-refinancing/>

Michael Neal, Linna Zhu, and Caitlin Young. "Why Automated Home Valuation Technology Errors Disproportionately Affect Majority-Black Neighborhoods," Urban Institute. May 5th, 2021. <https://www.urban.org/urban-wire/why-automated-home-valuation-technology-errors-disproportionately-affect-majority-black-neighborhoods>

RAND Fire Project:

Wallace, Rodrick, and Deborah Wallace. "Rand-HUD Fire Models." Management Science, vol. 26, no. 4, 1980, pp. 418–22, <https://www.jstor.org/stable/2630424>.

Jan Chaiken, Edward Ignall, Peter Kolesar, Warren Walker, (1980) Response to Communication on Rand-HUD Fire Models. Management Science 26(4):422-432.
<https://doi.org/10.1287/mnsc.26.4.422>.

Bonus Topics

TBD